

Recruitment Analytics

Project Overview:

This project focuses on analyzing recruitment data to evaluate hiring efficiency, identify bottlenecks in the hiring pipeline, and support data-driven decision-making. The dashboard provides a comprehensive view of candidate flow, recruiter performance, sourcing effectiveness, placements, and overall recruitment trends using Microsoft Power BI and MySQL.

Objectives

- Analyze recruitment data to understand hiring trends and recruitment performance.
- Track candidate movement across different stages of the hiring pipeline.
- Evaluate recruiter performance based on submissions, interviews, and joining's.
- Identify bottlenecks affecting the recruitment process and hiring efficiency.
- Measure important KPIs such as joining's, revenue, conversion rate and time-to-hire.
- Build an interactive dashboard to support data-driven recruitment decisions.

Dataset:

The dataset is structured across three Excel sheets

Job_Requirements_Table:

- Job_ID
- Role
- Client
- Priority
- Employment_Type
- Open_Date
- Close_Date
- TAT
- Requisition_Status

Candidate_Pipeline_Table:

- Candidate_ID
- Job_ID
- Visa_Status
- Source (LinkedIn, Dice, Job Boards, Referral, Internal Database)
- Recruiter_Name
- Submission_Date
- Interviewed
- Offer
- Joining_Status
- Backout
- TAT_Days

Financial_Table:

- Candidate_ID
- Job_ID
- Client
- Bill_Rate
- Pay_Rate
- Total_Margin

The three tables are linked relationally via Job_ID (between Jobs and Pipeline) and Candidate_ID (between Pipeline and Finance), forming a star-schema-like structure ideal for both SQL joins and Power BI data modelling.

Tools & Technologies:

- **Microsoft Excel** — Used for storing and handling raw recruitment data.
- **Python** — Used for data cleaning, preprocessing, and exploratory data analysis (EDA) using Pandas.
- **MySQL** — Used for database creation, applying primary key and foreign key relationships, additional data cleaning, and solving business queries.
- **Microsoft Power BI** — Used for KPI creation, dashboard development, and interactive data visualization.

Key Features:

- End-to-end recruitment analytics workflow
- Data cleaning and preprocessing using Python and MySQL
- Exploratory Data Analysis (EDA) for recruitment insights
- Relational database design using primary and foreign keys
- SQL-based business query analysis
- KPI tracking for placements, revenue, recruiter performance, and hiring efficiency
- Interactive dashboard creation using Microsoft Power BI
- Visual analysis of recruitment pipeline and sourcing effectiveness
- Business-focused insights for data-driven hiring decisions

Key Insights:

- Identified top-performing recruiters based on candidate closures and joining's.
- Analyzed candidate movement across different hiring stages to detect recruitment bottlenecks.
- Evaluated sourcing channels to understand the most effective candidate sources.
- Tracked hiring trends, joining success rate, and recruiter productivity.
- Measured important KPIs such as total joining's, revenue generated, and average time-to-hire.
- Developed interactive visualizations to support faster and data-driven recruitment decisions using Microsoft Power BI.

Data Cleaning:

In Excel:

- Structured raw data into three separate sheets (Jobs, Pipeline, Finance)
- Standardised column naming conventions across all sheets
- Ensured date columns used consistent Excel date format

In Python (Pandas):

- Loaded all three sheets using `pd.read_excel()` with `sheet_name` parameter
- Performed `shape`, `info()`, `describe()` checks on all three DataFrames
- Standardised all column names to lowercase using `.str.lower()`
- Detected and removed special characters in object columns using regex
- Handled missing values in `interview_date`, `offer_date`, `joining_date`, `tat_days`
- Converted date columns from object to datetime using `pd.to_datetime()`

In MySQL:

- Converted all date columns from VARCHAR to DATE using ALTER TABLE and SET SQL_SAFE_UPDATES
- Replaced empty strings with NULL in date and numeric columns using UPDATE statements
- Added Primary Keys (`candidate_id`, `job_id`) and Foreign Keys to enforce referential
- Created dimension tables: recruiter and client with auto-increment IDs
- Built a `recruitment_summary` view joining all three tables for simplified analysis

Python (Pandas) – EDA & Analysis:

Python was used for Exploratory Data Analysis (EDA) using Pandas, Matplotlib, and Seaborn. The analysis covered:

Visualisations Bar charts, pie charts, heatmaps, and line plots for submission trends and recruiter performance

EDA Recruiter Performance:

```
# EDA Recruiter Performance

plt.figure(figsize=(10,6))
pipeline_df.groupby('recruiter_name')['closure_flag'] \
    .sum() \
    .sort_values(ascending=False) \
    .head(10) \
    .plot(kind='bar')

plt.title("Top Recruiters by Closures")
plt.ylabel("Closures")
plt.show()
```



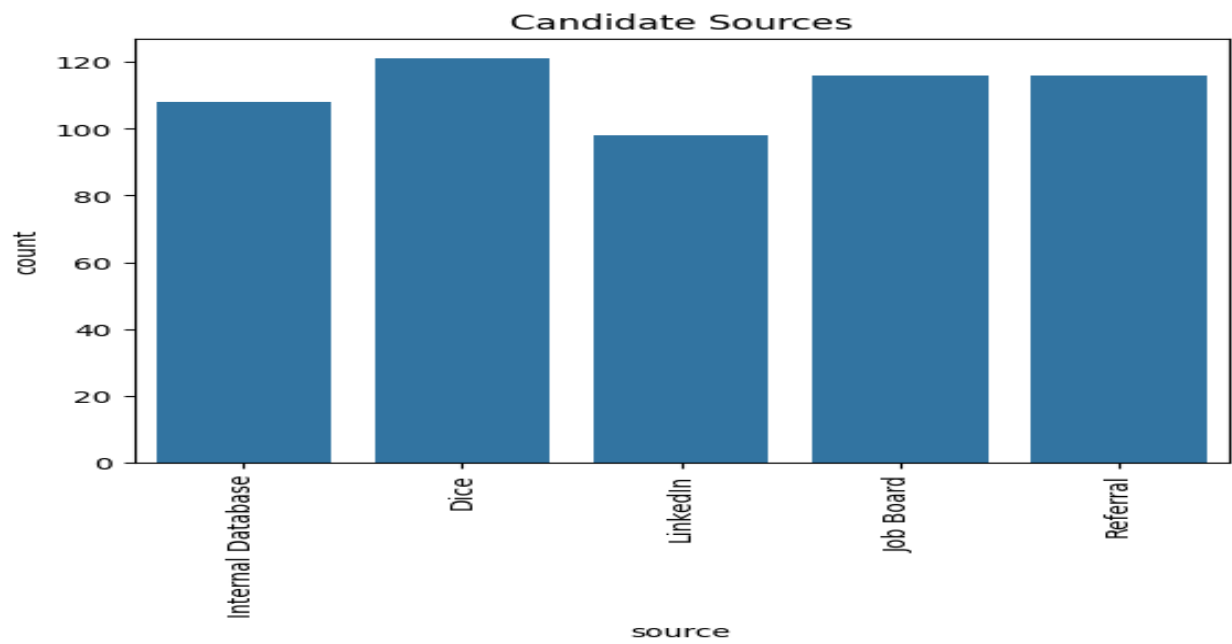
#Source Effectiveness

```
#Source Effectiveness

sns.countplot(
    data=pipeline_df,
    x='source'
)

plt.xticks(rotation=90)

plt.title("Candidate Sources")
plt.show()
```

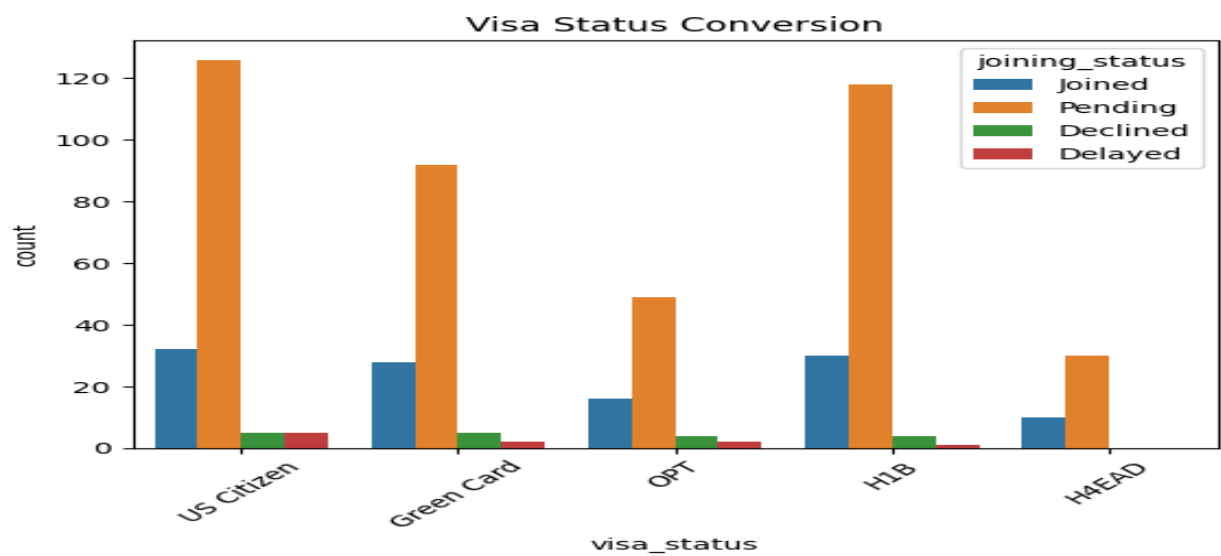


#Visa Status vs Joining:

```
#Visa Status vs Joining

sns.countplot(
    data=pipeline_df,
    x='visa_status',
    hue='joining_status'
)
plt.xticks(rotation=45)

plt.title("Visa Status Conversion")
plt.show()
```



SQL Analysis – Queries & Business Logic:

Pipeline Conversion Funnel : Counted total submissions, interviews, offers, and joins to measure funnel drop-off at each stage.

Recruiter Ranking by Closures & Margin: Used a CTE + DENSE_RANK window function to rank recruiters by total joins and revenue generated.

Backout Rate by Visa & Source: Calculated backout percentage grouped by visa_status and source to identify high-risk candidate segments.

Client-wise Fill Rate & Revenue: Measured fill rate (hires / total jobs) and total margin per client to assess client profitability.

Average TAT by Client & Priority: Analysed turnaround time variations by client and job priority to flag SLA breaches.

Source Effectiveness: Compared sourcing channels (Dice, LinkedIn, Referral, Job Board, Internal DB) by conversion rate and revenue.

Visa Status Conversion Rate: Compared hire-to-submission ratios across visa categories (H1B, GC, USC, OPT, H4EAD).

TAT Bucket Distribution: Categorised closed positions into Fast (<= 15 days), Moderate, Slow, and Critical (> 60 days) buckets.

Monthly Submission Trend: Tracked month-over-month submission volume and closure counts for trend analysis.

Notice Period Impact on Backout: Examined whether longer notice periods correlate with higher candidate backout rates.

Power BI Dashboard – Visualizations:

An interactive Power BI dashboard was built with two slicers (Client, Date) and the following visuals:

- KPI Cards — Total Submissions, Interviews, Offers, Joins, Revenue, Avg TAT, Conversion Rate, Total Clients
- Pipeline Conversion Funnel — Horizontal bar showing drop-off from Submissions → Interviews → Offers → Joins
- Recruiter Ranking — Bar chart showing closures and revenue per recruiter
- Quarterly Submissions Trend — Time series showing submission volume by quarter (2024–2026)
- Joining by Client — Donut chart with % share of joins per client
- Joinings by Visa — Horizontal bar comparing joins across H1B, GC, USC, OPT, H4EAD
- Source Effectiveness — Donut chart comparing sourcing channels
- Backout by Visa — Bar chart showing backout counts per visa category

Recommendations:

- Improve submission quality by strengthening pre-screening and improving JD-to-profile matching to reduce irrelevant submissions and increase interview conversion rates.
- Focus on high-performing recruiters such as Sanjay Mehta and Sunita Rao, who generate higher revenue per closure. Their sourcing methods and client engagement strategies can be analyzed and replicated across teams.
- Address aging job requisitions by reviewing positions that have remained open for a long period. Revising job requirements, prioritizing urgent roles, or closing inactive positions can improve recruitment efficiency.
- Improve client-specific hiring performance by analyzing low fill-rate accounts such as Wipro and identifying challenges related to salary expectations, role requirements, or interview processes.
- Optimize sourcing strategies by monitoring conversion rates across recruitment channels and allocating efforts toward higher-performing sources.
- Track visa-related risks more effectively by monitoring OPT candidate work authorization timelines to reduce last-stage drop-offs and hiring delays.

Conclusion:

This project successfully demonstrates an end-to-end Recruitment Analytics workflow using real-world US IT staffing data. The data was cleaned, analyzed, and transformed using Python and MySQL, followed by interactive dashboard creation in Microsoft Power BI.

The analysis provided valuable insights into recruiter performance, hiring trends, sourcing effectiveness, placements, and recruitment bottlenecks. By tracking important KPIs and visualizing recruitment data, the project supports data-driven decision-making and helps improve overall hiring efficiency.

This project also strengthened practical skills in data cleaning, SQL analysis, EDA, database design, and business intelligence reporting.

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